

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2024 Worked Solutions

October 2024 paper rebuilt with the worked solution after each question.

11

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 October 2024

October 2024 | 11 questions | 75 marks

11

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. A continuous curve has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	0.5	1.75	3	4.25	5.5
y	3.479	6.101	7.448	6.823	5.182

Using the trapezium rule with all the values of y in the given table,

- (a) find an estimate for

$$\int_{0.5}^{5.5} f(x) \, dx$$

giving your answer to one decimal place.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

$$\int_{0.5}^{5.5} (f(x) + 4x) \, dx$$

(2)

(Total for Question 1 is 5 marks)

(Total for Question 1 is 5 marks)

Worked Solution - Question 1

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: 90

Final answer

90

Question 2

Sequences & Series

2. A sequence of numbers $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where k is a constant.

- (a) Show that $u_5 = 7$

(3)

Given that $\sum_{r=1}^4 u_r = 30$

- (b) find the value of k .

(2)

- (c) Hence find the value of $\sum_{r=1}^{150} u_r$

(2)

(Total for Question 2 is 7 marks)

(Total for Question 2 is 7 marks)

Worked Solution - Question 2

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 3

Polynomials

3.

$$f(x) = 2x^3 - x^2 + Ax + B$$

where A and B are integers.

Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

(a) show that

$$3A - B = -118$$

(2)

Given also that $(2x - 5)$ is a factor of $f(x)$,

(b) find the value of A and the value of B .

(3)

(c) Hence find the quotient when $f(x)$ is divided by $(x - 7)$

(2)

(Total for Question 3 is 7 marks)

(Total for Question 3 is 7 marks)

Worked Solution - Question 3

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: $- + + + 2 \ 2 \ 13 \ 65 \ x \ x + +$

Final answer

$- + + + 2 \ 2 \ 13 \ 65 \ x \ x + +$

Question 4

Applications of Differentiation

4.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**The curve C has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ giving each term in simplest form. (2)
- (b) Hence find the x coordinate of the stationary point of C . (2)
- (c) (i) Find $\frac{d^2y}{dx^2}$ giving each term in simplest form.
- (ii) Hence determine the nature of the stationary point of C , giving a reason for your answer. (2)
- (d) State the range of values of x for which y is decreasing. (1)

(Total for Question 4 is 7 marks)

(Total for Question 4 is 7 marks)

Worked Solution - Question 4

Topic group

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: $y = -\frac{1}{2}x^2 + 2x + 0$ so (loc

Final answer

$y = -\frac{1}{2}x^2 + 2x + 0$ so (loc

Question 5

Binomial Expansion

5. (a) Find, in terms of a , the first 3 terms, in ascending powers of x , of the binomial expansion of

$$(2 + ax)^6$$

where a is a non-zero constant. Give each term in simplest form.

(3)

$$f(x) = \left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$$

Given that the constant term in the expansion of $f(x)$ is 576

- (b) find the value of a .

(4)

(Total for Question 5 is 7 marks)

(Total for Question 5 is 7 marks)

Worked Solution - Question 5

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: $- = 24 5 a = -$

Final answer

$- = 24 5 a = -$

Question 6

Laws of Logarithms

6.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Using the laws of logarithms, solve

$$\log_4(12 - 2x) = 2 + 2\log_4(x + 1)$$

(5)

(Total for Question 6 is 5 marks)

(Total for Question 6 is 5 marks)

Worked Solution - Question 6

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: $x = \dots$

Final answer

$$x = \frac{1}{8} \times 10^{\frac{1}{2}}$$

Question 7

Arithmetic Sequences

7. Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

- (a) Show that Jem pays £69.50 into scheme A in month 100 (1)
- (b) Find the **total** amount that Jem pays into scheme A over the period of 300 months. (2)

Kim pays money into a different savings scheme, B , over the same period of 300 months.

In a model, the amounts Kim pays into scheme B increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme B in month 1
 - £250 into scheme B in month 300
- (c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme A and the total amount paid into scheme B over the period of 300 months. (3)

(Total for Question 7 is 6 marks)

(Total for Question 7 is 6 marks)

Worked Solution - Question 7

1. Use the arithmetic formula

Identify the first term, common difference, and required term or sum from the question.

2. Substitute and solve

Use the arithmetic term or sum formula, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: = - 28425 27362

Final answer

= - 28425 27362

Question 8

Integration

8.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

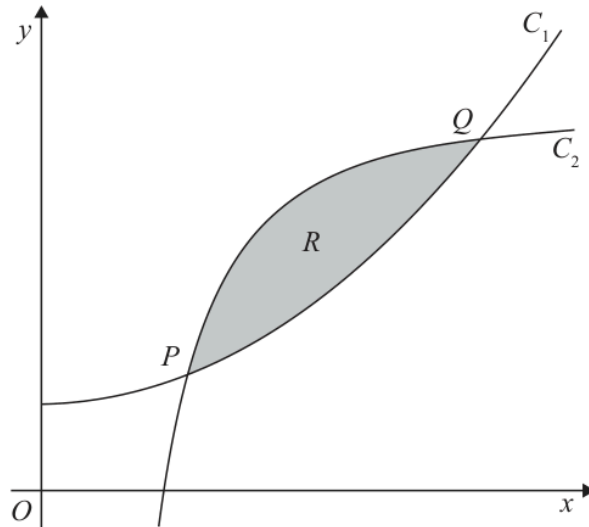


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation

$$y = x^2 + 3 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 13 - \frac{9}{x^2} \quad x > 0$$

The curves C_1 and C_2 intersect at the points P and Q as shown in Figure 1.(a) Use algebra to find the x coordinate of P and the x coordinate of Q .

(4)

The finite region R , shown shaded in Figure 1, is bounded by C_1 and C_2 (b) Use algebraic integration to find the exact area of R .

(4)

(Total for Question 8 is 8 marks)

(Total for Question 8 is 8 marks)

Worked Solution - Question 8

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $+ - + - + - + = 16 \ 3 =$

Final answer

$$+ - + - + - + = 16 \ 3 =$$

Question 9

Trigonometric Equations

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$2 \tan \theta = 3 \cos \theta$$

can be written as

$$3 \sin^2 \theta + 2 \sin \theta - 3 = 0$$

(3)

(b) Hence solve, for $-\pi < x < \pi$, the equation

$$2 \tan \left(2x + \frac{\pi}{3} \right) = 3 \cos \left(2x + \frac{\pi}{3} \right)$$

giving your answers to 3 significant figures.

(4)

(Total for Question 9 is 7 marks)

(Total for Question 9 is 7 marks)

Worked Solution - Question 9

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: $\theta = 0$

Final answer

$$\theta = 0$$

Question 10

Circles

10. The circle C has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact value of the radius of C .

(3)

The line L has equation $y = mx + 1$, where m is a constant.

Given that L is the tangent to C at the point P ,

(b) show that

$$2m^2 - 7m - 22 = 0$$

(3)

(c) Hence find the possible pairs of coordinates of P .

(4)

(Total for Question 10 is 10 marks)

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 11

Proof

11. (i) Prove by counter example that the statement

“If n is a prime number then $3^n + 2$ is also a prime number.”

is false.

(2)

(ii) Use proof by exhaustion to prove that if m is an integer that is **not** divisible by 3, then

$$m^2 - 1$$

is divisible by 3

(4)

(Total for Question 11 is 6 marks)

(Total for Question 11 is 6 marks)

Worked Solution - Question 11

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: not divisible by 3

Final answer

not divisible by 3